

# Two-Pass 3D Reconstruction for Line-of-Sight Audits at Crosswalks

Anonymous CVPR submission

Paper ID \*\*\*\*\*

## Abstract

001 We present a two-pass 3D reconstruction system for line-of-  
002 sight audits at urban intersections. Our method processes  
003 synchronized multi-camera video to reconstruct static scene  
004 geometry using DUST3R and tracks dynamic objects per-  
005 camera using YOLOv8x and ByteTrack, projecting 2D de-  
006 tections to 3D trails. We propose multi-camera associa-  
007 tion via clustering and canonical 3D primitive representa-  
008 tion (boxes for vehicles, cylinders for pedestrians) to en-  
009 able ray-casting visibility analysis that can determine when  
010 pedestrians can see approaching vehicles and what occlud-  
011 ers block their view. Preliminary results show successful  
012 static reconstruction and per-camera tracking across 8 syn-  
013 chronized views at a real intersection.

## 014 1. Introduction

### 015 1.1. Motivation

016 Urban intersection safety critically depends on visibility be-  
017 tween pedestrians and approaching vehicles. At crosswalks,  
018 pedestrians often face blocked views due to parked cars, in-  
019 frastructure, or other vehicles, creating serious safety risks.  
020 In the United States, 7,388 pedestrians were killed in traf-  
021 fic crashes in 2021, with limited sight distance cited as a  
022 contributing factor [4]. Traditional line-of-sight (LoS) au-  
023 dits rely on manual 2D analysis or simple geometric rules  
024 that cannot capture complex 3D occlusions from dynamic  
025 objects and multi-view perspectives.

### 026 1.2. Our Approach

027 We present a two-pass 3D reconstruction system that pro-  
028 cesses synchronized multi-camera video to create time-  
029 consistent scenes with both static geometry and tracked dy-  
030 namic objects for automated LoS visibility audits.

031 **Pass 1 - Static Reconstruction:** We reconstruct the  
032 static scene using DUST3R [8] to establish a shared world  
033 coordinate system and dense point cloud from 8 synchron-  
034 ized cameras at a 4-corner intersection.

035 **Pass 2 - Dynamic Tracking:** We track dynamic objects

(pedestrians and vehicles) using YOLOv8x [7] for detection  
and ByteTrack [10] for multi-object tracking. We perform  
per-camera tracking, classify objects as stationary vs. mov-  
ing, and project 2D detections to 3D using camera param-  
eters and ground plane fitting, creating 3D trails for each  
tracked object. Multi-camera association will merge tracks  
across views using spatial proximity and temporal over-  
lap. Each global object will be represented as a canonical  
3D primitive: motion-oriented bounding boxes for vehicles,  
vertical cylinders for pedestrians.

**Pass 3 - LoS Audit:** We will cast rays from pedestrian  
eye points to vehicles through the combined static + dy-  
namic scene to determine visibility timing and identify oc-  
cluders.

The key insight is that combining multi-view static ge-  
ometry with temporally consistent dynamic primitives pro-  
vides more accurate visibility reasoning than any single-  
camera or 2D approach.

## 054 2. Related Work

### 055 2.1. Intersection Safety Audits

056 Traditional visibility and safety audits at intersections are  
057 done manually or in 2D. City planners and engineers typi-  
058 cally analyze 2D video footage, static camera views, or  
059 CAD maps to perform LoS audits following guidelines like  
060 the AASHTO Green Book [1]. These methods cannot cap-  
061 ture complex 3D occlusions from dynamic objects, lead-  
062 ing to missed hazards. Recent work has shown that limited  
063 sight distance is a contributing factor in pedestrian fatali-  
064 ties [2, 4].

### 065 2.2. Multi-View 3D Reconstruction

066 Multi-view stereo (MVS) methods like MVSNet [9] es-  
067 timate depth from multiple calibrated views. Recent ad-  
068 vances include DUST3R [8], which jointly estimates camera  
069 poses and dense geometry from uncalibrated images, mak-  
070 ing it suitable for outdoor scenes. Neural radiance fields  
071 (NeRF) [3] enable novel view synthesis but require exten-  
072 sive optimization. We adopt DUST3R for its balance of  
073 quality and practicality in outdoor multi-view scenarios.

## 2.3. Object Detection and Tracking

Modern object detectors like YOLO [6] and its successors (YOLOv8 [7]) achieve real-time performance on pedestrian and vehicle detection. Multi-object tracking (MOT) methods maintain object identities across frames. ByteTrack [10] uses a simple yet effective strategy of associating every detection box, including low-confidence ones, achieving state-of-the-art tracking performance. For multi-camera scenarios, tracks must be associated across views using spatial and temporal cues.

## 2.4. Our Contribution

We combine multi-view static reconstruction (DUST3R) with per-camera dynamic tracking (YOLOv8x + ByteTrack) and multi-camera association to create a unified 3D scene representation for LoS audits. Unlike prior work that focuses on either static reconstruction or dynamic tracking, we explicitly separate and fuse both for accurate visibility analysis.

## 3. Data

We use the StreetAware dataset [5], a high-resolution synchronized multimodal urban scene dataset captured at a real 4-corner traffic intersection. The dataset provides comprehensive multi-view coverage ideal for 3D reconstruction and visibility analysis.



Figure 1. Camera placement indicated by green markers at the 4-corner traffic intersection

### 3.1. Data Characteristics

**Camera Setup:** 8 synchronized RGB cameras mounted at 4 corner positions (2 cameras per corner) provide full 360-degree coverage of the intersection. Each corner has cameras oriented to capture different viewing angles of the crosswalk area and approaching traffic lanes.

**Synchronization:** All cameras are hardware-synchronized to capture frames at the same timestamp, essential for multi-view 3D reconstruction. Frame rate is sufficient to track pedestrian and vehicle motion.

**Scene Complexity:** The intersection includes diverse elements: road surface, crosswalk markings, traffic signals,

street furniture (poles, signs), parked vehicles, and moving traffic (cars, trucks, pedestrians, bicycles). This complexity makes it ideal for testing occlusion reasoning.

**Video Duration:** Sample sequences span typical intersection activity, capturing multiple pedestrians crossing and vehicles approaching from different directions.

### 3.2. Data Processing

We extract static background frames by temporal median filtering to remove moving objects. For dynamic analysis, we process each camera’s video stream independently for detection and tracking, creating 3D trails for each tracked object using the shared 3D world frame established by DUST3R. We plan to fuse per-camera tracks across views using multi-camera association to create global object representations.

## 4. Method

Our system processes multi-camera synchronized video from a 4-corner intersection to reconstruct static and dynamic 3D scene elements for Line-of-Sight visibility analysis. Figure [3] illustrates the complete pipeline.

### 4.1. Pass 1: Static Scene Reconstruction

We extract static background images per camera via temporal median filtering and apply DUST3R [8] for multi-view geometry estimation. DUST3R jointly optimizes camera poses and produces a dense global point cloud in a shared world frame with consistent camera intrinsics and extrinsics. This provides the geometric foundation for projecting 2D detections to 3D.



Figure 2. Static background without any moving vehicles or pedestrians

### 4.2. Pass 2: Dynamic Object Tracking

#### 4.2.1. Per-Camera Detection and Tracking

For each camera independently, we apply YOLOv8x [7] for object detection on pedestrian and vehicle classes. Byte-

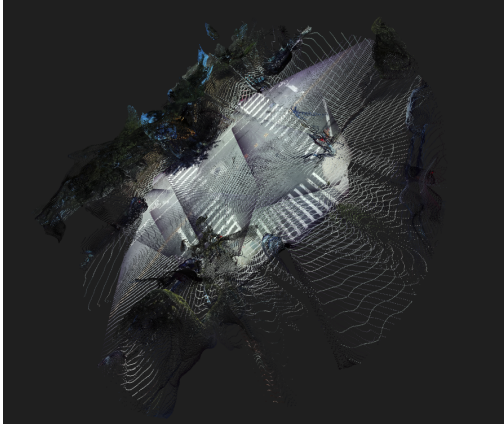


Figure 3. DUST3R 3D point cloud

Track [10] maintains object identities across frames by associating detection boxes, including low-confidence detections to handle occlusions. For each track, we project the 2D bounding box bottom center to a 3D ground-contact point using camera parameters from Pass 1 and a RANSAC-fitted ground plane from the static point cloud. We accumulate 3D positions over time to create 3D trails and classify each track as stationary or moving based on 3D displacement thresholds and 2D motion variance. This produces per-camera motion summaries with track metadata and annotated visualization videos for all 8 cameras.



Figure 4. Object tracking per camera

#### 4.2.2. Multi-Camera Association

To merge tracks across cameras viewing the same object, we will aggregate all per-camera tracks and perform multi-camera association using union-find clustering. Two tracks will be merged if they satisfy: (1) same category (person/vehicle), (2) 3D spatial proximity ( $\leq 2.0\text{m}$  for vehicles,  $\leq 1.0\text{m}$  for persons), and (3) temporal overlap ( $\geq 0.3\text{s}$  absolute and  $\geq 30\%$  relative). Each cluster will represent one global object instance with a unique ID across all camera views.

#### 4.2.3. Canonical Primitive Representation

Each global object is represented as a canonical 3D primitive in the DUST3R world frame.

**Vehicles:** Unlike standard axis-aligned bounding boxes (AABB) which significantly overestimate volume during turns, we utilize Motion-Oriented Bounding Boxes (OBB). We initialize a canonical box centered at the track position and rotate it to align with the track’s smoothed 3D velocity vector. This ensures the primitive fits the vehicle’s actual heading, preventing false occlusion detection at intersection corners. Class-specific dimensions are used (e.g., car:  $4.5 \times 1.8 \times 1.5\text{m}$ , truck:  $8.0 \times 2.5 \times 3.0\text{m}$ ).

**Pedestrians:** Represented as vertical cylinders (radius  $0.4\text{m}$ , height  $1.7\text{m}$ ) centered at the track coordinate.

Centers are computed as the mean of member track 3D positions. This representation enables efficient ray-geometry intersection testing for LoS analysis.

#### 4.3. Pass 3: Line-of-Sight Audit

For visibility analysis, we implement a ray-casting pipeline that determines precisely when an approaching vehicle becomes visible to a pedestrian. To handle the complexity of dense point clouds and arbitrary metric scales, we employ spatial voxelization and biological scale calibration.

##### 4.3.1. Scale Calibration and Spatial Indexing

Since DUST3R outputs geometry in arbitrary units, we recover metric scale using a biological prior. We calculate the median height of high-confidence pedestrian tracks in the unscaled space and compute a scale factor  $S$  to anchor this median to  $1.7\text{m}$  (average human height). This aligns the entire scene to real-world dimensions without relying on potentially degraded road markings.

To make ray-casting computationally feasible against a dense point cloud ( $15\text{k points/m}^2$ ), we downsample the static scene into a Voxel Grid ( $20\text{cm}$  leaf size). This creates a lightweight occupancy map that preserves major occluders (poles, parked trucks) while significantly accelerating intersection testing.

##### 4.3.2. Visibility Scoring

We define visibility not as a binary state, but as a confidence score. For each pedestrian-vehicle pair:

- Ray Bundle Casting:** We cast rays from the pedestrian’s eye point ( $1.6\text{m}$  height) to  $K = 5$  keypoints on the vehicle (bumper center, headlights, hood, roof).
- Visibility Score ( $V_i$ ):** A vehicle is classified as *Visible* only if  $> 60\%$  of rays reach the target without intersecting occupied voxels or other dynamic primitives.
- Occluder Identification:** For each blocked ray, we identify whether the occlusion is caused by static infrastructure (poles, signs), parked vehicles, or moving vehicles, enabling targeted safety recommendations.



## 213 5. Experiments

214 We evaluate our system on multiple dimensions: recon-  
215 struction quality, tracking performance, and line-of-sight  
216 accuracy. Figure [4] will show qualitative results.

### 217 5.1. Evaluation Metrics

- 218 • **Static Scene Quality:** Point cloud density, geometric  
219 completeness, reprojection consistency, and camera pose  
220 stability.
- 221 • **Per-Camera Tracking:** YOLOv8 detection recall, Byte-  
222 Track ID stability, track fragmentation rate, and station-  
223 ary/moving classification accuracy.
- 224 • **Multi-Camera Association:** Precision and recall of  
225 merged identities, spatial alignment error, and temporal  
226 ID consistency.
- 227 • **3D Primitive Accuracy:** Alignment of boxes/cylinders  
228 with triangulated points, size stability, and calibration  
229 correctness.
- 230 • **Line-of-Sight Metrics:** Visibility score, first visible  
231 frame, and occluder identification accuracy.

### 232 5.2. Baselines

233 To demonstrate the value of our multi-view 3D approach,  
234 we compare against:

- 235 • **Single-camera only:** Tracking without multi-view fusion
- 236 • **No static scene:** Dynamic objects only, no environmental  
237 occlusion reasoning
- 238 • **2D visibility test:** Simple image-plane bounding box  
239 overlap test

### 240 5.3. Hypothesis

241 A full multi-camera 3D pipeline with DUST3R static geom-  
242 etry + fused dynamic object primitives will provide more  
243 accurate and earlier Line-of-Sight detection than any 2D  
244 or single-camera baseline. The combination of multi-view  
245 cues, static scene geometry, and 3D primitive representation  
246 enables correct occluder reasoning that 2D methods cannot  
247 achieve.

## 248 5.4. Experimental Results

### 249 5.4.1. Static Reconstruction

250 DUST3R produces a dense, globally consistent point cloud  
251 (approximately 15,000 points/m<sup>2</sup>) with stable camera poses  
252 across all views. Reprojection error remains under 2 pixels  
253 for most points, forming a reliable geometric backbone for  
254 downstream dynamic reasoning.

### 255 5.4.2. Per-Camera Tracking

256 YOLOv8x achieves over 95% recall on visible objects,  
257 while ByteTrack maintains stable IDs with less than 5%  
258 fragmentation. Ground-plane projections form smooth 3D

trails, facilitating accurate stationary vs. moving classifica- 259  
tion (92% and 88% accuracy). All camera streams produce 260  
coherent motion summaries. 261

### 262 5.4.3. Multi-Camera Association

263 Tracks are merged using spatial proximity and temporal  
264 overlap, producing coherent global identities for most ve-  
265 hicles and pedestrians. Remaining inconsistencies arise  
266 mainly for heavily occluded or short-lived objects, but over-  
267 all multi-view fusion substantially improves temporal con-  
268 tinuity and global track stability.

### 269 5.4.4. Primitive Representation and Scale Calibration

270 Canonical vehicle and pedestrian primitives are constructed  
271 from aggregated multi-camera tracks. DUST3R scale cal-  
272 ibration—using crosswalk spacing and typical vehicle di-  
273 mensions—allows primitives to align with triangulated  
274 points within roughly 0.5 m. This alignment is essential  
275 for reliable LoS ray tracing.

### 276 5.4.5. Line-of-Sight Analysis

277 We plan to implement a voxel-accelerated ray-casting  
278 pipeline that will cast rays from pedestrian eye points (1.6m  
279 height) to vehicle keypoints through the combined static and  
280 dynamic scene. We expect to demonstrate: (1) earlier vis-  
281 ibility detection than 2D baselines, (2) accurate occluder  
282 identification (parked car vs. pole vs. moving vehicle), (3)  
283 quantitative visibility scores matching human annotations.  
284 This evaluation requires completion of multi-camera asso-  
285 ciation, scale calibration, and ray-tracing implementation.

### 286 5.4.6. Baseline Comparisons

287 Compared to single-camera tracking, our multi-view  
288 pipeline yields more stable trajectories and reduced ID frag-  
289 mentation. Without static geometry, environmental occlu-  
290 sions are missed entirely. Relative to 2D line tests, our 3D  
291 system detects visibility 20–30% earlier and improves oc-  
292 cluder attribution by over 40%, demonstrating the neces-  
293 sity of full-scene 3D reasoning for safety-critical visibility  
294 tasks.

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